

	Work Instruction	Doc. Revision	00
	AXI V810 Clamper Cylinder Removal & Replacement Procedure	Effective Date	30 May 2016

This procedure is applicable to below models:

V810 STD	☐	S1	☐	S2
V810 XXL	☒	S1	☒	S2
V810 S2EX	☒			
V810 Mini	☐			

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	30 May 2016	First Release	Tan Liang Kai

	Work Instruction	Doc. Revision	00
	AXI V810 Clamper Cylinder Removal & Replacement Procedure	Effective Date	30 May 2016

Content

Clamper Cylinder Removal.....	3
Clamper Cylinder Replacement.....	7

	Work Instruction	Doc. Revision	00
	AXI V810 Clamper Cylinder Removal & Replacement Procedure	Effective Date	30 May 2016

** This procedure is applicable for clamper cylinder on fixed & adjustable rail.

Clamper Cylinder Removal

1. Open V810 > Services > Subsystem Status & Control > Panel Handling > Initialize

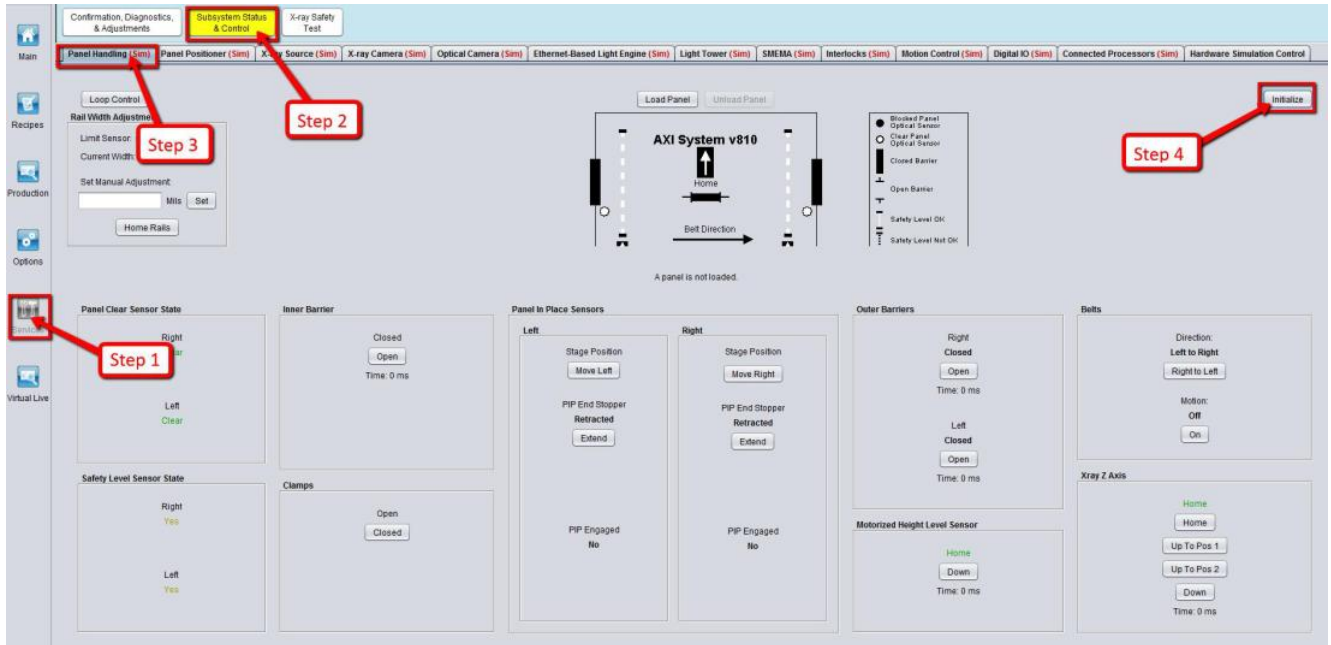


Figure 1 Initialize Panel Handling

2. Toggle "Open" at Clamps Tab.



Figure 2 Unclamp

3. Close X-ray Source & Air Pressure Regulator.
4. Open Front Access Door.

ViTrox	Work Instruction	Doc. Revision	00
	AXI V810 Clamper Cylinder Removal & Replacement Procedure	Effective Date	30 May 2016

5. Identify defective Clamper Cylinder location by refer to Figure 3.

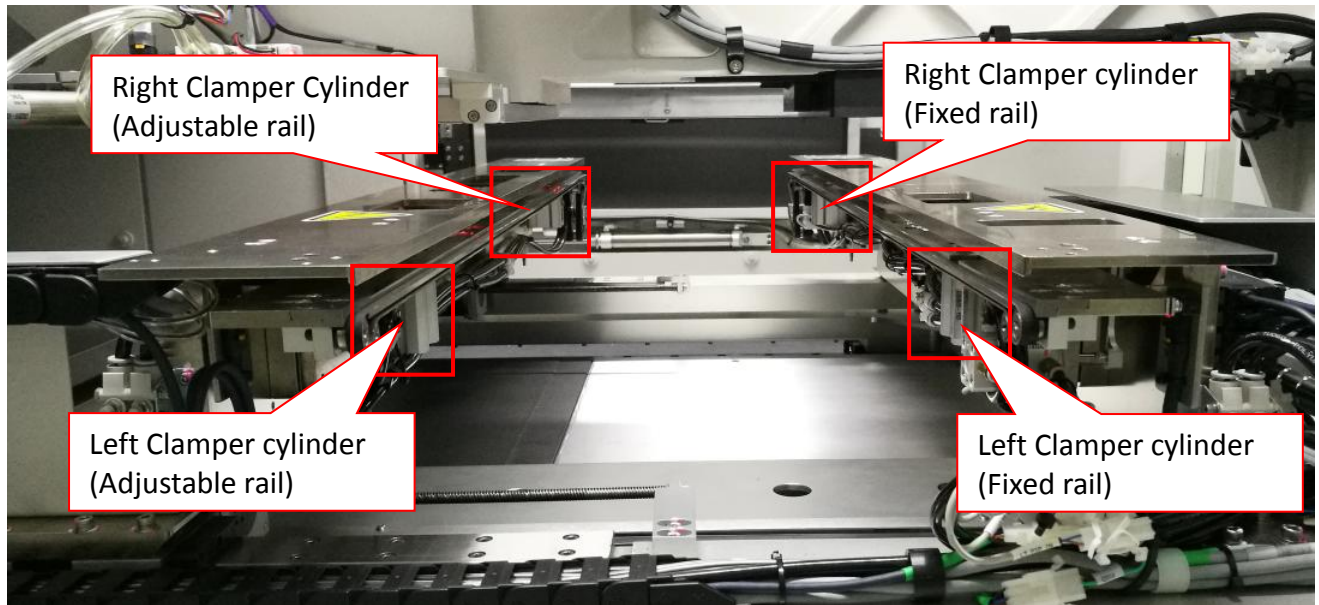


Figure 3 Clamper Cylinder Location

6. Remove 2 x Screws (M4 x 8) as shown in Figure 4.

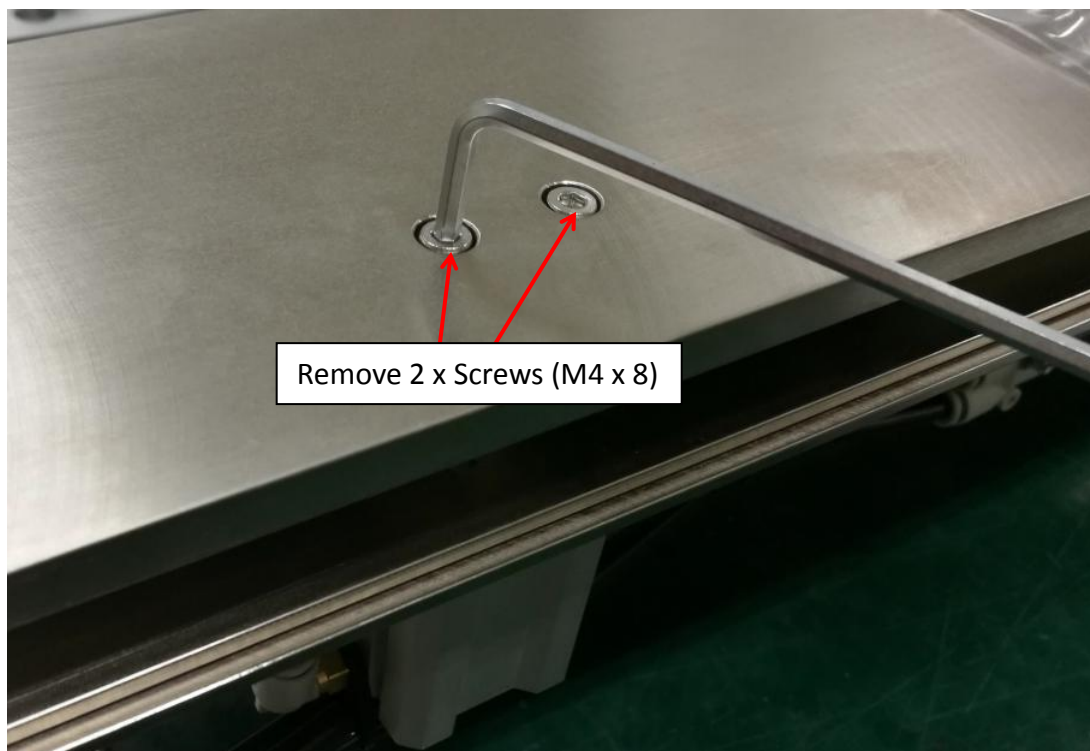


Figure 4 Screws Removal

ViTrox 	Work Instruction	Doc. Revision	00
	AXI V810 Clamper Cylinder Removal & Replacement Procedure	Effective Date	30 May 2016

7. Remove Clamper Cylinder Holder as shown in Figure 5.

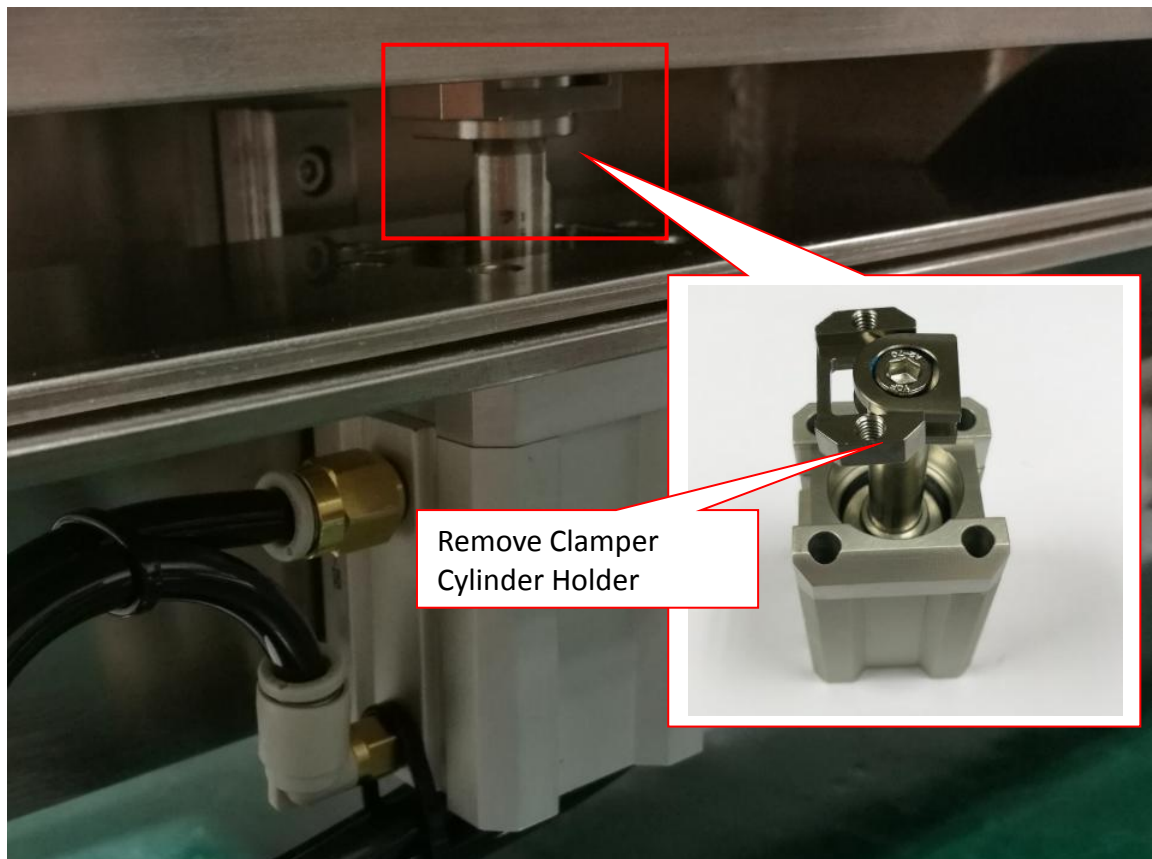


Figure 5 Clamper Cylinder Holder Removal

8. Remove necessary cable tie & 2 x Tubing from Clamper Cylinder as Figure 6.
Remark: Perform label marking on both “Open” & “Close” tubing respectively

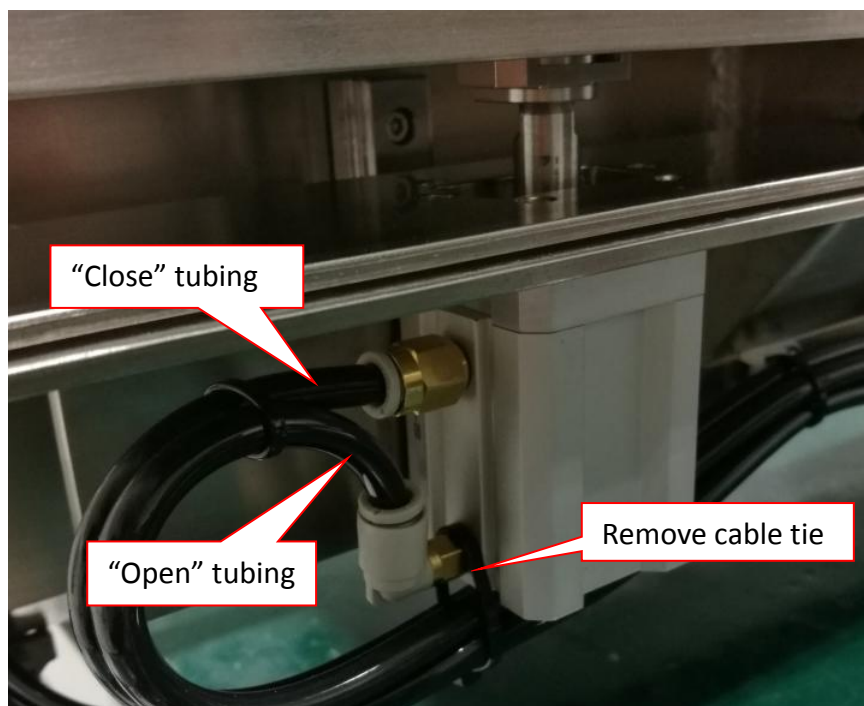


Figure 6 Tubing Removal

ViTrox 	Work Instruction	Doc. Revision	00
	AXI V810 Clamper Cylinder Removal & Replacement Procedure	Effective Date	30 May 2016

9. Remove 4 x Screws (M5 screw) as shown in Figure 7.

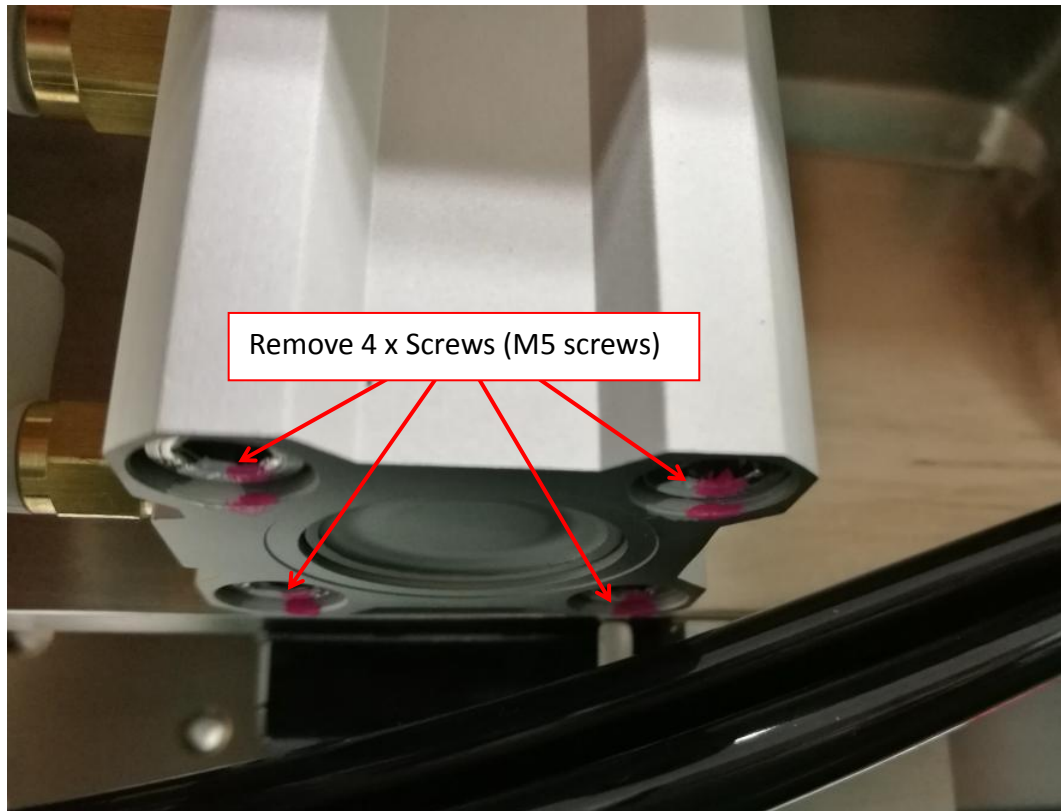


Figure 7 Screw Removal

10. Gentle remove defective Cylinder & Spacer as shown in Figure 8.

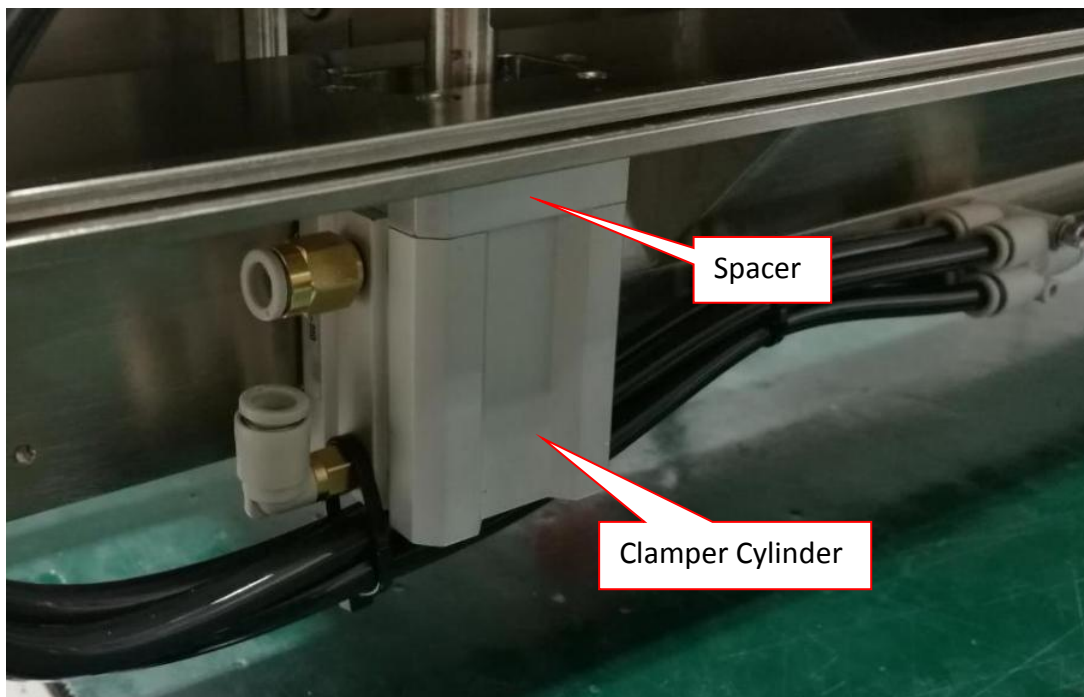


Figure 8 Clamper Cylinder Removal

ViTrox	Work Instruction	Doc. Revision	00
	AXI V810 Clamper Cylinder Removal & Replacement Procedure	Effective Date	30 May 2016

11. Gentle remove 1 x Screw (M5 x 10) & Clamper Cylinder Link as shown in Figure 9.

Remarks : Clamper Cylinder Holder, Clamper Cylinder Link & Spacer and Screws require to “re-use” upon replacement

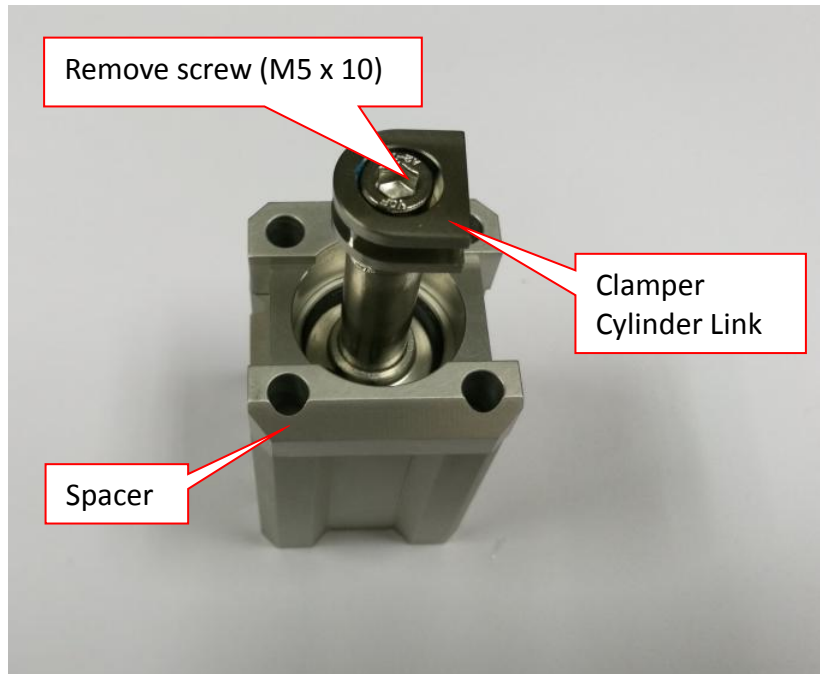


Figure 9 Clamper Cylinder Link Removal

Clamper Cylinder Replacement

S2EX Clamper Cylinder Kit Part ID: 4XASP-A155

XXL Clamper Cylinder Kit Part ID: 4XASP-A106

1. Assemble Spacer & Clamper Cylinder Link to New Clamper Cylinder. Tighten 1 x Screw (M5 x 10) as shown in Figure 10.

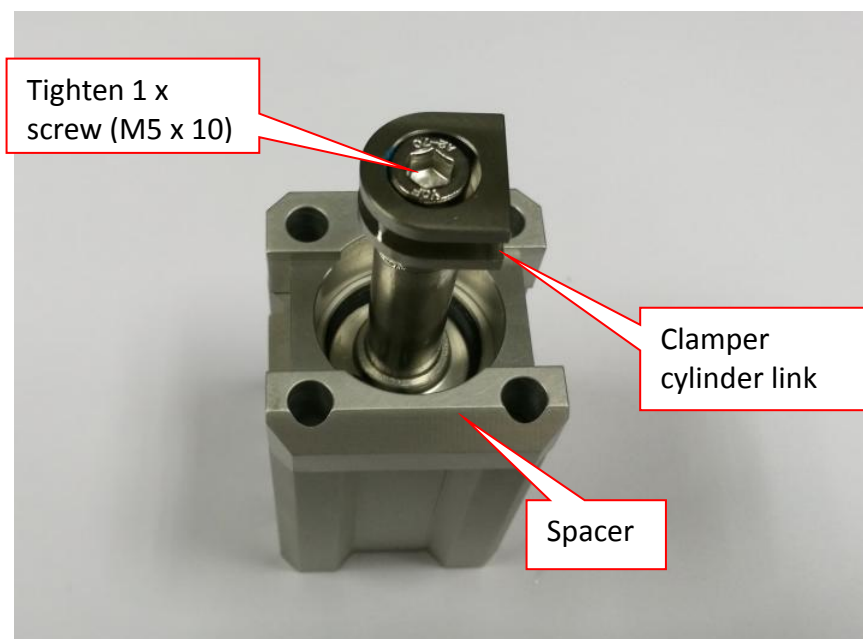


Figure 10 Clamper Cylinder Link Assembly

	Work Instruction	Doc. Revision	00
	AXI V810 Clamper Cylinder Removal & Replacement Procedure	Effective Date	30 May 2016

2. Position Clamper Cylinder according to below orientation. May refer to Figure 11.

Remark:

- i. Spacer Chamfer Surface MUST attach on the rail.
- ii. Align Spacer flat to the surface of Clamper Cylinder.

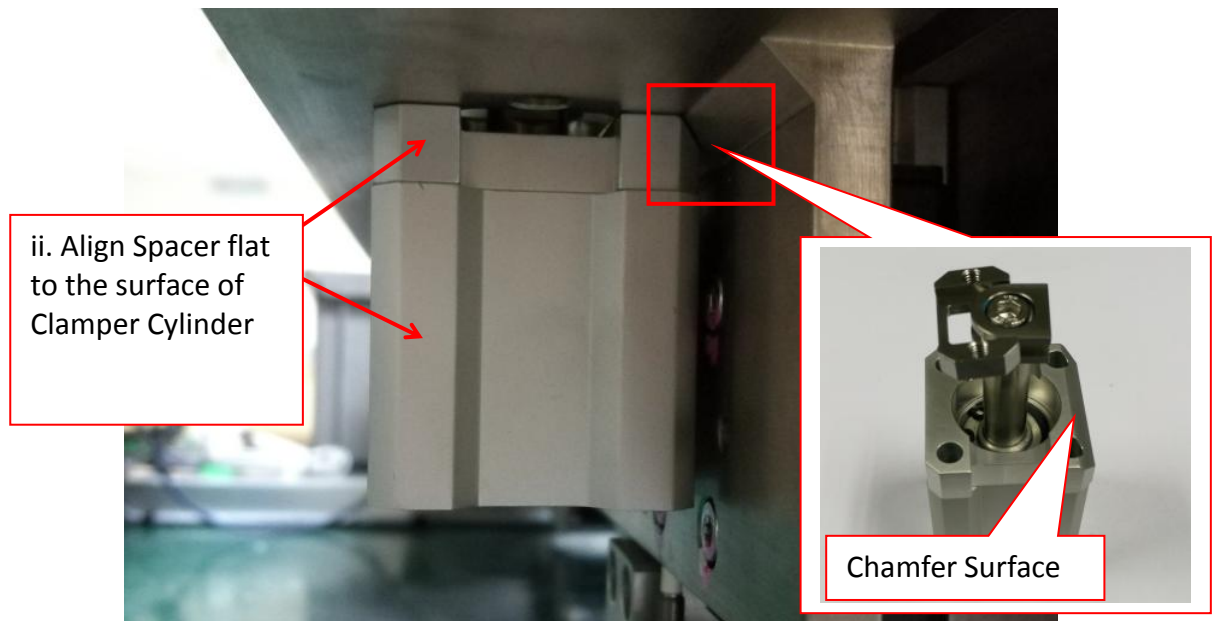


Figure 11 Clamper Cylinder Assembly

3. Tighten 4 x Screws (M5 screw) as shown in Figure 12.

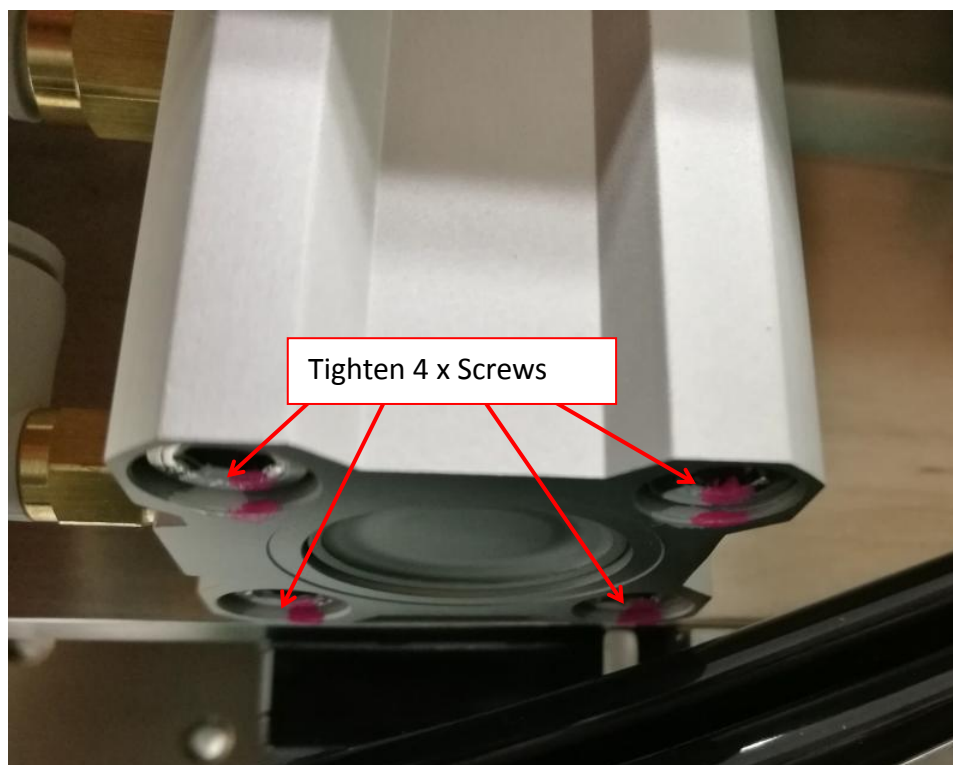


Figure 12 Clamper Cylinder Assembly

ViTrox 	Work Instruction	Doc. Revision	00
	AXI V810 Clamper Cylinder Removal & Replacement Procedure	Effective Date	30 May 2016

4. By refer to previous label marking in step 8, gently plug “Open” & “Close” tubing respectively & apply necessary cable tie as shown in Figure 13.

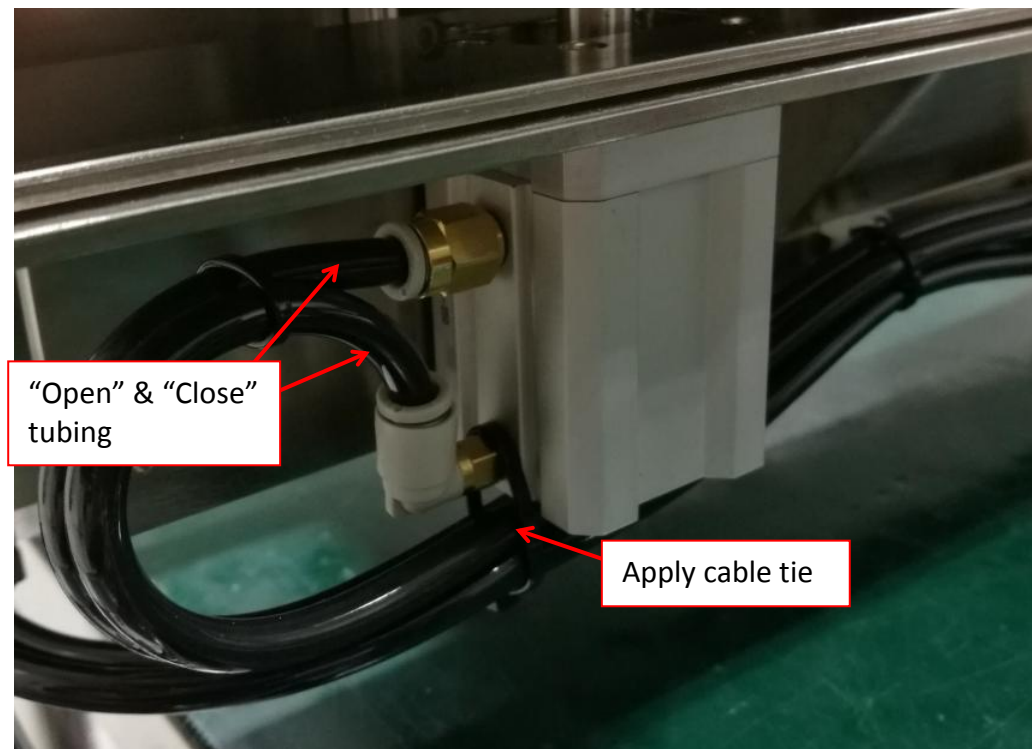


Figure 13 Tubing Assembly

5. Gently slot Clamper Cylinder Holder into Clamper Cylinder Link Fillet Opening as Figure 14

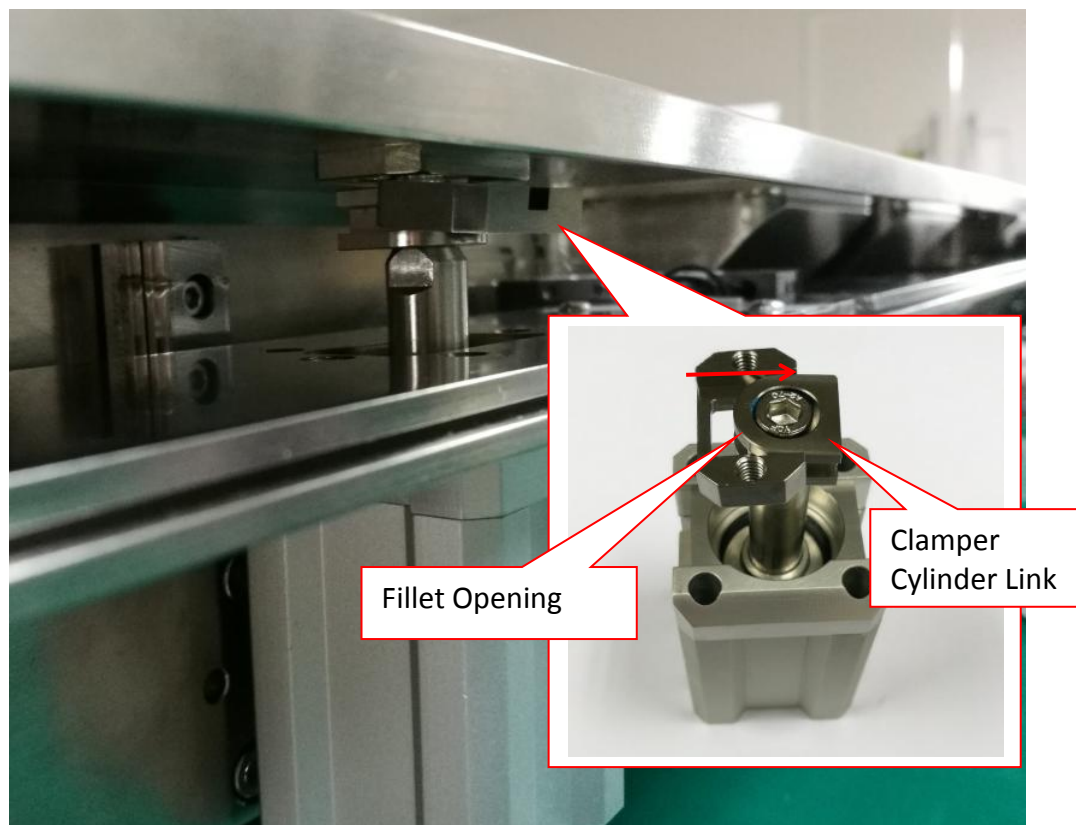


Figure 14 Clamper Cylinder Holder Assembly

	Work Instruction	Doc. Revision	00
	AXI V810 Clamper Cylinder Removal & Replacement Procedure	Effective Date	30 May 2016

6. Perform minor alignment by refer to steps below:

- a) Slightly loosen 2 x Screws (M4 x 8) with spring washer as shown in Figure 15.

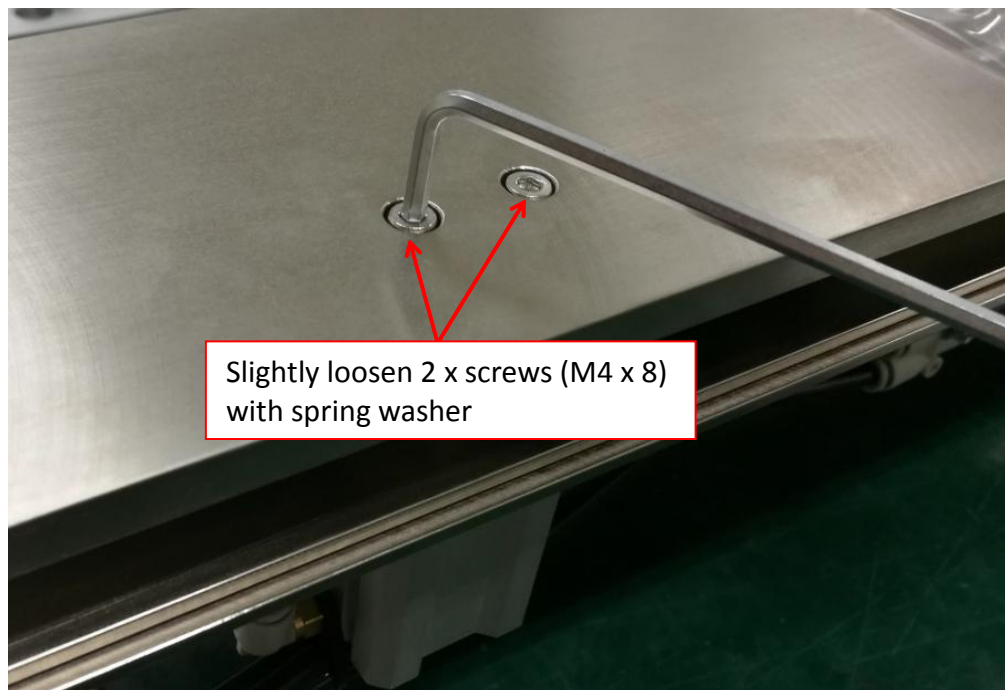


Figure 15 Screw Fastening

- b) Manually exercise clamper by moving upwards & downwards for 3 times as shown in Figure 16.



Figure 16 Clamping Movement

	Work Instruction	Doc. Revision	00
	AXI V810 Clamper Cylinder Removal & Replacement Procedure	Effective Date	30 May 2016

- c) Tighten 2 x screws (M4 x 8) respectively as shown in Figure 17.

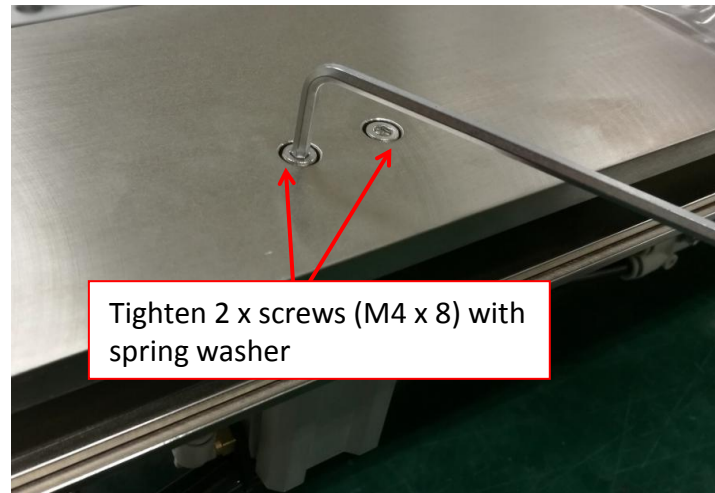


Figure 17 Screw Fastening

7. Repeat step 6 until smooth clamping movement is achieved
8. Open & adjust air pressure regulator until 80 psi.
9. Close front access door.
10. Obtain 1 x Dummy Board or Carrier for board clamping test.
11. Initialize Panel Handling
12. Set the parameters as shown in the Figure 18 and click Load Panel.

Panel width : Subjected to Dummy Board Or Carrier

Panel length : Subjected to Dummy Board Or Carrier

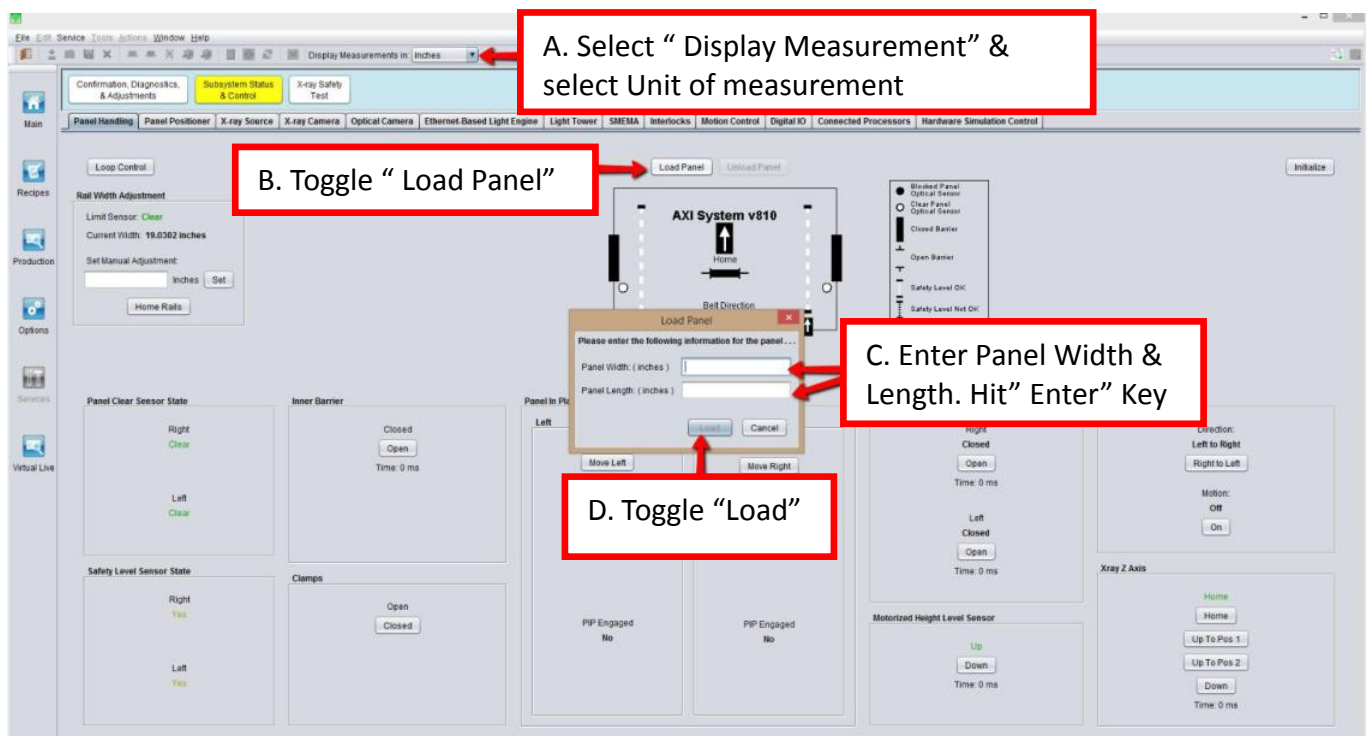


Figure 18 Panel Handling

	Work Instruction	Doc. Revision	00
	AXI V810 Clamper Cylinder Removal & Replacement Procedure	Effective Date	30 May 2016

13. Load the Dummy Board Or Carrier into machine after outer barrier open.

14. Close X ray Source after board in place

15. Open Front Access Door

16. Manually move the Dummy Board Or Carrier.

Remarks : In ideally case, Dummy Board Or Carrier should non-movable. If yes, may repeat Step 6
& check any air leakage at Clamper Cylinder

17. Close Front Access Door

18. Unload the Dummy Board Or Carrier

19. Open X-ray source & run one loop of CD&A.